# Solubility

CHEMISTRY REFERENCE

| cations ↓ \ anions → | $NO_3^-$ nitrate | $C_2H_3O_2^-$ acetate | $ClO_3^-$ chlorate | $Cl^-$ chloride | $Br^-$ bromide | $I^-$ iodide | $SO_4^{2-}$ sulfate | $OH^-$ hydroxide | $CO_3^{2-}$ carbonate | $PO_4^{3-}$ phosphate | $S^{2-}$ sulfide |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $NH_4^+$ ammonium | S | S | S | S | S | S | S | S | S | S | S |
| $Na^+$ sodium | S | S | S | S | S | S | S | S | S | S | S |
| $K^+$ potassium | S | S | S | S | S | S | S | S | S | S | S |
| $Mg^{2+}$ magnesium | S | S | S | S | S | S | S | I | I | I | X |
| $Ca^{2+}$ calcium | S | S | S | S | S | S | *SS* | *SS* | I | I | X |
| $Sr^{2+}$ strontium | S | S | S | S | S | S | I | *SS* | I | I | X |
| $Ba^{2+}$ barium | S | S | S | S | S | S | I | S | I | I | X |
| $Al^{3+}$ aluminum | S | S | S | S | S | S | S | I | X | I | X |
| $Mn^{2+}$ manganese(II) | S | S | S | S | S | S | S | I | I | I | I |
| $Fe^{2+}$ iron(II) | S | S | S | S | S | S | S | I | I | I | I |
| $Fe^{3+}$ iron(III) | S | S | S | S | S | X | S | I | X | I | X |
| $Co^{2+}$ cobalt(II) | S | S | S | S | S | S | S | I | I | I | I |
| $Ni^{2+}$ nickel(II) | S | S | S | S | S | S | S | I | I | I | I |
| $Cu^{2+}$ copper(II) | S | S | S | S | S | X | S | I | I | I | I |
| $Zn^{2+}$ zinc | S | S | S | S | S | S | S | I | I | I | I |
| $Ag^+$ silver | S | *SS* | S | I | I | I | *SS* | I | I | I | I |
| $Hg_2^{2+}$ mercury(I) | S | *SS* | S | I | I | I | I | I | I | I | I |
| $Pb^{2+}$ lead(II) | S | S | S | *SS* | *SS* | I | I | I | I | I | I |

**S** soluble (more than 1 g per 100 mL)
**SS** slightly soluble (0.1–1 g)
**I** insoluble (less than 0.1 g) — a precipitate
**X** reacts with water or does not form
Water at about 25 °C.

## Rules

1. Compounds of Group 1 ions and $NH_4^+$ are soluble. So are all nitrates and chlorates, and nearly all acetates.
2. Chlorides, bromides and iodides are soluble, except with $Ag^+$, $Hg_2^{2+}$ and $Pb^{2+}$.
3. Sulfates are soluble, except with $Sr^{2+}$, $Ba^{2+}$, $Pb^{2+}$ and $Hg_2^{2+}$. $Ca^{2+}$ and $Ag^+$ sulfates are slightly soluble.
4. Hydroxides are insoluble, except with Group 1, $NH_4^+$ and $Ba^{2+}$. $Ca^{2+}$ and $Sr^{2+}$ hydroxides are slightly soluble.
5. Carbonates, phosphates and sulfides are insoluble, except with Group 1 and $NH_4^+$. Group 2 sulfides react with water.
6. **X** cells break these rules on purpose: those ions react with each other or with water instead of forming the compound. Check the grid.